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Ring polymer molecular dynamics with surface hopping

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We propose a ring polymer molecular dynamics method for the calculation of chemical rate constants that incorporates nonadiabatic effects by the surface-hopping approach. Two approximate ring polymer electronic Hamiltonians are formulated and the time-dependent Schrodinger equation for the electronic amplitudes is solved self-consistently with the ring polymer equations of motion. The beads of the ring polymer move on a single adiabatic potential energy surface at all times except for instantaneous surface hops. The probability for a hop is determined by the fewest-switches surface-hopping criterion. During a surface hop all beads switch simultaneously to the new potential energy surface with positions kept unchanged and momenta adjusted properly to conserve total energy. The approach allows the evaluation of total rate coefficients as well as electronic state-selected contributions. The method is tested against exact quantum mechanical calculations for a one-dimensional, two-state model system that mimics a prototypical nonadiabatic bimolecular chemical reaction. For this model system, the method reproduces quite accurately the tunneling contribution to the rate and the distribution of reactants between the electronic states. © *2012 American Institute of Physics*. [http://dx.doi.org/10.1063/1.4766449]

I. INTRODUCTION

The computation of accurate chemical reaction rates poses a considerable challenge in chemical physics. Significant progress has been made by utilizing classical molecular dynamics simulations, which allow calculations on systems containing large numbers of particles. The molecular dynamics approach is based on the classical propagation of nuclei on a single potential energy surface defined by a unique nondegenerate state of the electronic subsystem and as such it suffers from a few fundamental constraints. Quantum mechanical effects such as tunneling and zero-point energy can play an important role, particularly for processes taking place at low temperature and involving light particles. In addition, in regions of configuration space where potential energy surfaces cross or become close to each other, the Born-Oppenheimer approximation[1] may break down and electronic transitions can occur. Unfortunately, the exact quantum mechanical treatment of medium to large-scale chemical systems is a formidable computational problem. Therefore, approximate approaches that improve on the classical mechanical framework are of special practical interest.

Over the past decades, a variety of approximate quantum dynamical methods have emerged that include tunneling and zero-point energy in the calculation of rate coefficients.[2–13] Two of these approaches, ring polymer molecular dynamics (RPMD)[4–6]and centroid molecular dynamics (CMD)[7–9]are based on the imaginary-time path-integral formalism,[14,15] in which the quantum Boltzmann distribution is mapped exactly on a classical system of ring polymers. RPMD employs this correspondence to approximate the exact quantum evolution with the classical mechanical propagation of the ring polymers. The method has been applied to a number of chemical reactions in condensed[16–18] and gas phases,[19,20] where it generally provides satisfactory results. In some of these applications,[17,18] RPMD has been utilized to simulate nonadiabatic processes in the framework of a mean-field-like one-electron pseudopotential approximation, where a single effective electron represented by a necklace of beads models the totality of the electronic degrees of freedom. However, this approach cannot properly describe entanglement, i.e., nuclear motion evolving differently on each potential energy surface.

Nonadiabatic effects have been successfully incorporated in the classical dynamics of the nuclei by a number of methods.[21–34] The fewest-switches surface-hopping (FSSH) approach[22] has become one of the most widely used techniques due to its reliable performance and ease of implementation. In this method, the amplitudes of the quantum degrees of freedom are propagated self-consistently and simultaneously with the trajectory of the classical degrees of freedom. The surface-hopping probability is computed on-the-fly and ensures proper distribution of a swarm of trajectories among the quantum states consistent with the quantum populations. A limitation of the method is the assumption that classical mechanics provides an adequate description of the nuclear motion. Although this is a reliable assumption in many situations, there are cases in which certain aspects of the nuclear quantum behavior become important. For instance, a reaction of substantial interest in chemical physics is proton-coupled electron transfer,[35,36] where interplay of quantum transitions and proton tunneling determines the reaction mechanism and the rate coefficients. The FSSH method can be applied to such phenomena by properly augmenting the quantum subsystem with all relevant degrees of freedom that exhibit

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quantum behavior.[37,38] Nevertheless, nuclei are generally massive enough that quantum coherences tend to wash out, especially in condensed phase systems,[39] and the quantum phenomena that survive are quantum statistical effects such as tunneling and zero-point energy. RPMD is specifically tailored to address this kind of situation.[4]

The objective of this paper is to present a molecular dynamics method for the calculation of reaction rate constants that accurately incorporates nonadiabatic and quantum statistical effects. The method is essentially an approximation to the full quantum dynamics that has favorable scalability with system size such that it can be applied to realistic multidimensional systems. The approach is developed by blending ideas from FSSH and RPMD. However, we have not provided a rigorous derivation of the new method, but rather we have tried to justify it on the basis of physical arguments.

The paper is organized as follows: in Sec. II, the quantum-mechanical rate theory is reviewed; the classical rate theory together with its extension to nonadiabatic processes is discussed briefly in Sec. III; and a short review of adiabatic RPMD in Sec. IV concludes the introductory part of the paper. Section V presents the new method and an efficient computational algorithm is outlined in Sec. VI. The new method is applied to a two-state one-dimensional model system and the performance is tested against accurate quantum-mechanical results in Sec. VII. The computational details are provided in the Appendix. In Sec. VIII, the main features and assumptions of the new method are discussed. Throughout the paper, the energy is measured in units of frequency such that $\hbar$ is set to unity.

II. QUANTUM-MECHANICAL RATE THEORY

In Secs. II–VII, we consider a quantum-mechanical system composed of electrons and nuclei. For simplicity, we focus our discussion on a single nuclear degree of freedom with mass M, i.e., the "reaction coordinate." The generalization to multiple dimensions and many nuclei with different masses is straightforward.

Let $\mathbf{r}$ designate the collection of electronic coordinates and R stand for the nuclear position. The Hamiltonian of the combined electronic and nuclear motion is

$$\hat{H}(\mathbf{r}, R) = \hat{T}(R) + \hat{H}_{el}(\mathbf{r}; R), \tag{1}$$

where $\hat{T}(R)$ is the nuclear kinetic energy operator, and $\hat{H}_{el}(\mathbf{r}; R)$ is the electronic Hamiltonian operator at fixed nuclear position.

Let us divide the nuclear configuration space into two regions, A and B that belong to the reactants and the products of a chemical reaction, respectively. The separation is achieved by a dividing surface $s(R) = 0$ such that $s(R) < 0$ corresponds to the reactant region and $s(R) > 0$ locates the product region. In one-dimensional space, the dividing surface is simply a point located at $R^{\neq}$ and $s(R) = R - R^{\neq}$. Following linear response theory,[40,41] the quantum-mechanical (QM) rate coefficient is given by the ratio of the long-time limit of the Kubo-transformed flux-side correlation function $C_{fs}(T, t)$ and the reactant partition function Q_A,

$$k_{\mathrm{QM}} = \frac{1}{Q_A} \lim_{t\to\infty} C_{fs}(T, t), \tag{2}$$

where T is temperature, and t is time. The flux-side correlation function is given by[42]

$$C_{fs}(T, t) = \int_0^\beta \frac{d\lambda}{\beta} tr[e^{-(\beta-\lambda)\hat{H}} \hat{F} e^{-\lambda\hat{H}} e^{i\hat{H}t} \hat{h} e^{-i\hat{H}t}]. \tag{3}$$

Here, $\beta = 1/k_B T$ is the reciprocal temperature and $\hat{h} = h(s(R))$ is the projection operator on the product side of the dividing surface with h the Heaviside step function ($h(s < 0) = 0$ and $h(s > 0) = 1$). The flux operator is the Heisenberg time derivative of the projection operator $\hat{F} = i[\hat{H}, \hat{h}]$ and the trace in Eq. (3) is taken over both the electronic and nuclear degrees of freedom.

The electrons are usually described in an adiabatic basis defined by the eigenvalue equation

$$\hat{H}_{el}(\mathbf{r}; R)\varphi_i(\mathbf{r}; R) = V_i(R)\varphi_i(\mathbf{r}; R). \tag{4}$$

The adiabatic eigenfunctions $\varphi_i(\mathbf{r}; R)$ constitute an orthonormal basis at each nuclear position and the eigenvalues $V_i(R)$ are the adiabatic potential energy surfaces for nuclear motion. The adiabatic basis functions at different nuclear positions are not necessarily orthogonal and, as the nuclei move with finite momentum, the nuclear motion on different potential energy surfaces couples.

The adiabatic approximation[1] results from the neglect of the coupling among the potential energy surfaces and in many chemical reactions the nuclear motion can be adequately described as taking place solely on the ground adiabatic potential energy surface (PES). For such chemical reactions, only the term corresponding to the ground PES survives after taking the trace over the electronic degrees of freedom and the flux-side correlation function in Eq. (3) takes the form

$$C_{fs}(T, t) = \int_0^\beta \frac{d\lambda}{\beta} tr[e^{-(\beta-\lambda)\hat{H}^{ad}} \hat{F} e^{-\lambda\hat{H}^{ad}} e^{i\hat{H}^{ad}t} \hat{h} e^{-i\hat{H}^{ad}t}] \tag{5}$$

with trace only over the nuclear coordinates and $\hat{H}^{ad}$ the adiabatic Hamiltonian

$$\hat{H}^{ad} = \hat{T}(R) + \hat{V}_0(R). \tag{6}$$

In the same approximation, the reactant partition function in Eq. (2) for a chemical reaction that occurs in the gas phase becomes the translational partition function per unit length

$$Q_A = \left(\frac{M}{2\pi\beta}\right)^{\frac{1}{2}}. \tag{7}$$

III. CLASSICAL RATE THEORY

Initially, we assume that the nuclear motion is classical throughout and discuss briefly how nonadiabatic effects can be incorporated in the calculation of chemical rate coefficients.

A. Adiabatic classical rate theory

In the classical limit, the trace over the nuclear degrees of freedom becomes integration over phase space and functions of the phase space variables are substituted for quantum operators. The adiabatic Hamiltonian of Eq. (6) then becomes

$$H^{ad}(R, P) = \frac{P^2}{2M} + V_0(R), \tag{8}$$

where P is the nuclear momentum. Further, the classical rate coefficient is given by

$$k_{cl} = \frac{1}{Q_A^{cl}} \lim_{t\to\infty} C_{fs}^{cl}(T, t). \tag{9}$$

Here, Q_A^{cl} is the classical reactant partition function per unit length and

$$C_{fs}^{cl}(T, t) = \frac{1}{2\pi} \iint dR_0\, dP_0 e^{-\beta H^{ad}(R_0, P_0)} \times \delta(R_0 - R^{\neq}) \frac{P_0}{M} h(R_t - R^{\neq}) \tag{10}$$

is the classical flux-side correlation function. The flux through the dividing surface at $t = 0$ is defined as

$$F(R, P) = \frac{dh(R_0 - R^{\neq})}{dt} = \delta(R_0 - R^{\neq}) \frac{P_0}{M}. \tag{11}$$

Therefore, in the classical limit the computation of the chemical rate involves equilibrium sampling of initial positions R_0 and momenta P_0 at the dividing surface followed by propagation of classical trajectories R_t, P_t. The initial momenta that contribute to the reactive flux give rise to trajectories that end up in the product region after long enough time of propagation. These reactive trajectories are selected by the projection operator, $h(R_t - R^{\neq})$ in Eq. (10).

The fraction of nonreactive trajectories can be significantly minimized by an appropriate choice of the dividing surface. For activated chemical processes, the trajectories that connect reactants with products usually proceed through a bottleneck region, where a large portion of the trajectories evolve to the product side. The reactant molecules access this region with low thermal probability and the rate constant can be calculated efficiently if the dividing surface is located in the bottleneck region, which is also called the transition state.[43–45] The rate coefficient can be written as a product of two contributions

$$k_{cl} = k_{\mathrm{TST}}^{cl} \kappa^{cl}. \tag{12}$$

The first term is the classical transition state theory (TST) rate coefficient given by

$$k_{\mathrm{TST}}^{cl} = \frac{1}{Q_A^{cl}} \lim_{t\to 0_+} C_{fs}^{cl}(T, t). \tag{13}$$

Here, $C_{fs}^{cl}(T, 0_+)$ is the short-time limit of the classical flux-side correlation function

$$C_{fs}^{cl}(T, 0_+) = \frac{1}{2\pi} \iint dR_0\, dP_0 e^{-\beta H^{ad}(R_0, P_0)} \times \delta(R_0 - R^{\neq}) \frac{P_0}{M} h(P_0) \tag{14}$$

and represents the outward thermal flux through the dividing surface. The transmission coefficient or recrossing correction, the second term on the rhs of Eq. (12), takes care of the reduction of the thermal flux due to recrossing of the dividing surface and is described by the ratio

$$\kappa^{cl} = \frac{\lim_{t\to\infty} C_{fs}^{cl}(T, t)}{\lim_{t\to 0_+} C_{fs}^{cl}(T, t)}. \tag{15}$$

The classical TST rate coefficient is an upper limit of the total rate coefficient and it can be used for the variational optimization of the location of the dividing surface, which reduces substantially the fraction of nonreactive trajectories in the computation of the transmission coefficient.

B. Fewest-switches surface-hopping

The extension of classical rate theory to nonadiabatic processes requires the incorporation of nonadiabatic transitions in the molecular dynamics approach.[22] For this purpose, we assume that the nuclei move along some classical trajectory $R(t)$. The electronic Hamiltonian then becomes parametrically dependent on time through the nuclear trajectory. Further, the time-dependent electronic wave function $\psi(\mathbf{r}, R, t)$ can be expanded in the adiabatic basis, $\varphi_i(\mathbf{r}; R)$, as follows:

$$\psi(\mathbf{r}, R, t) = \sum_i c_i(t) \varphi_i(\mathbf{r}; R), \tag{16}$$

where $c_i(t)$ are the time-dependent complex expansion coefficients. In this basis, the time-dependent Schrodinger equation for the electrons becomes

$$i\dot{c}_i = V_i(R) c_i - i \sum_j \dot{R} \cdot d_{ij}(R) c_j. \tag{17}$$

The first term on the rhs of Eq. (17) describes the usual adiabatic phase evolution with $V_i(R)$ defined in Eq. (4) and the second term is responsible for transfer of electronic amplitude between different adiabatic states. The rate of transfer is proportional to the dot product of the nuclear velocity and the nonadiabatic coupling vector defined as

$$d_{ij} = \langle \varphi_i(\mathbf{r}; R) | \nabla_R | \varphi_j(\mathbf{r}; R) \rangle. \tag{18}$$

The brackets denote integration over the electronic coordinates only. We solve Eq. (17) numerically along the nuclear trajectory to obtain the time evolution of the electronic amplitudes.

In the surface-hopping methods,[23–26] the nuclei move on a single potential energy surface except for instantaneous hops. In particular, the FSSH approach apportions trajectories among the electronic states according to the quantum populations $|c_i(t)|^2$. The probability for switching from the current state i to some other state j in the time interval from t to $t + \Delta t$ is

$$g_{ij} = \frac{\Delta t\, b_{ji}}{a_{ii}}, \tag{19}$$

where $a_{ij} = c_i c_j^*$ and $b_{ji} = -2\mathrm{Re}(a_{ji}^* \dot{R} \cdot d_{ji})$. If g_{ij} is negative, then it is set to zero. A uniform random number between zero and one is then selected to determine whether a switch

to state j will take place. If no switch occurs, then the nuclear motion continues on the current PES. If a switch to state j takes place and $V_i(R) \neq V_j(R)$, then the nuclei begin to evolve on the V_j PES and the momentum parallel to the nonadiabatic coupling vector is adjusted to conserve the total energy. If the required momentum reduction is greater than the component of momentum along the nonadiabatic coupling vector, then the state switch is rejected.

The switching procedure ensures that except for rejected hops, for a large ensemble of trajectories the portion of trajectories on a given state at any time t equals the quantum populations.

C. Nonadiabatic classical rate theory

As proposed in Ref. 46, the adiabatic classical rate theory can be extended in a straightforward manner to nonadiabatic chemical reactions with the help of the FSSH method. We place the dividing surface far in the reactant region, where we assume that the nonadiabatic coupling vanishes. The equilibrium distributions of initial positions and momenta at this dividing surface are sampled independently on each PES and a swarm of trajectories (R_t), representing the thermal flux out of the dividing surface, is propagated forward in time by the FSSH algorithm of Sec. III B, which takes into account the nonadiabatic coupling among the PESs along the way to the product region. With these definitions in mind, the mixed quantum-classical (MQC) flux-side correlation function becomes

$$C_{fs}^{\mathrm{MQC}}(T,t) = \frac{1}{2\pi}\sum_i \iint dR_0\, dP_0 e^{-\beta H^{ad}(R_0,P_0;i)} \times \delta(R_0 - R^{\neq})\frac{P_0}{M} h(R_t - R^{\neq}) \tag{20}$$

with

$$H^{ad}(R,P;i) = \frac{P^2}{2M} + V_i(R). \tag{21}$$

As described above, the subscript i labels the adiabatic PES and the summation is over all electronic states, which results from the trace over the electronic degrees of freedom in Eq. (3). The dividing surface is independent of the electronic quantum number and is common to all potential energy surfaces. The rate coefficient is similarly given by

$$k_{\mathrm{MQC}} = \frac{1}{Q_A^{cl}} \lim_{t\to\infty} C_{fs}^{\mathrm{MQC}}(T,t), \tag{22}$$

where the reactant partition function per unit volume receives additional contribution from the thermal electronic populations in the asymptotic reactant region

$$Q_A^{cl} = \left(\frac{M}{2\pi\beta}\right)^{\frac{1}{2}} \lim_{R\to-\infty} \sum_i e^{-\beta V_i(R)}. \tag{23}$$

In cases where reactive chemical events are infrequent, significant computational advantage can be gained if the present approach for rate coefficient calculation is made independent of the location of the dividing surface. To approximately take this into account, trajectories are propagated forward and backward in time from the dividing surface together with an appropriate reweighting procedure. The details of the rare event FSSH algorithm are summarized in Sec. VI.

The short-time limit of the rate coefficient in Eq. (22) is simply the adiabatic TST rate generalized to multiple potential energy surfaces (MPES),

$$k_{\mathrm{MPES-TST}}^{cl} = \frac{1}{Q_A^{cl}} \sum_i C_{fs,i}^{cl}(T,0_+), \tag{24}$$

where $C_{fs,i}^{cl}(T,0_+)$ is again the short-time limit of the classical correlation function on each adiabatic PES,

$$C_{fs,i}^{cl}(T,0_+) = \frac{1}{2\pi}\iint dR_0\, dP_0 e^{-\beta H^{ad}(R_0,P_0;i)} \times \delta(R_0 - R^{\neq})\frac{P_0}{M} h(P_0). \tag{25}$$

Note that the nonadiabatic effects are included in the recrossing of the dividing surface. Furthermore, the transmission coefficient can be split into contributions from different electronic states in the asymptotic reactant and product regions. We define the state-selected transmission coefficient that represents the reactive flux of trajectories starting on electronic state *in* in the distant past, crossing the dividing surface on state i, and ending up on state *fi* in the distant future as

$$\kappa_{in,fi}^{i} = \frac{\lim_{t\to\infty} \iint dR_0 dP_0\, \delta_{in,n(-t)}\, e^{-\beta H^{ad}(R_0,P_0;i)} \delta(R_0 - R^{\neq})\frac{P_0}{M} h(R_t - R^{\neq})\, \delta_{fi,n(t)}}{C_{fs,i}^{cl}(T,0_+)}. \tag{26}$$

Here, $n(t)$ is the electronic state of a nuclear trajectory at time t and $\delta_{m,n}$ is the Kronecker delta function. With the help of Eqs. (24) and (26), the MQC rate coefficient of Eq. (22) can be rewritten as

$$k_{\mathrm{MQC}} = \frac{1}{Q_A^{cl}} \sum_{in,fi}\sum_i C_{fs,i}^{cl}(T,0_+)\kappa_{in,fi}^{i}. \tag{27}$$

Performing the sum over i, we obtain

$$k_{\mathrm{MQC}} = \sum_{in,fi} k_{in,fi}^{\mathrm{MQC}}, \tag{28}$$

i.e., the total MQC rate is the sum of the initial and final state-selected thermal rate coefficients with obvious notation.

IV. ADIABATIC RING POLYMER RATE THEORY

In RPMD,[4] each quantum particle is mapped on a necklace of beads. The beads move on a single adiabatic PES and adjacent beads interact via a temperature-dependent harmonic potential. The classical RPMD Hamiltonian that corresponds to the quantum-mechanical Hamiltonian operator is

$$H_N^{ad} = \sum_{k=1}^{N} \frac{P_k^2}{2M} + \sum_{k=1}^{N} \frac{M}{2}\omega_N^2 (R_{k+1} - R_k)^2 + \sum_{k=1}^{N} V(R_k) \tag{29}$$

with P_k and R_k the bead momenta and positions, respectively, and $R_{N+1} = R_1$. The mass of each bead equals the mass of the quantum particle and the frequency of the harmonic interaction is $\omega_N = 1/\beta_N$. The number of beads in a ring polymer is N, and $\beta_N = \beta/N$.

The RPMD rate theory[5,6] is based on the following approximation to the flux-side Kubo-transformed correlation function in Eq. (5):

$$C_{fs}^N(T,t) = \frac{1}{(2\pi)^N} \iint d\mathbf{R}_0\, d\mathbf{P}_0 e^{-\beta_N H_N^{ad}(\mathbf{R}_0,\mathbf{P}_0)} \times \delta(\bar{R}_0 - R^{\neq}) \frac{\bar{P}_0}{M} h(\bar{R}_t - R^{\neq}), \tag{30}$$

where the beads evolve according to the classical equations of motion generated by the RPMD Hamiltonian in Eq. (29). In Eq. (30), $\bar{R}$ and $\bar{P}$ are the centroid position and momentum, respectively, defined as the ring polymer center-of-mass position and momentum

$$\bar{R} = \frac{1}{N}\sum_{k=1}^{N} R_k, \tag{31}$$

$$\bar{P} = \frac{1}{N}\sum_{k=1}^{N} P_k. \tag{32}$$

From Eq. (2), the RPMD rate coefficient is given by

$$k_{\mathrm{RPMD}} = \frac{1}{Q_A^N} \lim_{t\to\infty} C_{fs}^N(T,t), \tag{33}$$

and Q_A^N is the N-bead representation of the reactant partition function per unit length.

The RPMD rate coefficient is similar to the classical rate coefficient, Eqs. (9) and (10), with the position of the dividing surface depending only on the centroid coordinate. Essentially, RPMD rate theory is a classical rate theory in the extended phase space of the N-bead ring polymer. A special place is reserved for the centroid variables, which play the role of the position and momentum of a classical-like particle; however, the centroid trajectory is obtained by the propagation of the whole ring polymer.

RPMD rate theory has a number of valuable features. It provides the exact quantum-mechanical rate coefficient in the shallow-tunneling regime,[5] where the reaction barrier can be approximated by a parabolic potential, and it gives an adequate approximation to the semiclassical instanton rate in the deep-tunneling region.[47] Moreover, the RPMD rate coefficient reduces to the classical rate expression in the high-temperature region, where the number of beads can be set to one ($N = 1$). Equations (30) and (33) have a well defined short-time limit,[48] which coincides with the centroid version of quantum transition state theory (QTST),[11–13]

$$k_{\mathrm{QTST}} = \frac{1}{Q_A^N} \iint d\mathbf{R}_0\, d\mathbf{P}_0 e^{-\beta_N H_N^{ad}(\mathbf{R}_0,\mathbf{P}_0)} \delta(\bar{R}_0 - R^{\neq}) \frac{\bar{P}_0}{M} h(\bar{P}_0). \tag{34}$$

The QTST rate is an upper limit to the RPMD rate and can be used to determine variationally the most appropriate location of the dividing surface. While the QTST is a valuable tool in the calculation of the total RPMD rate coefficient, a significant computational advantage of RPMD rate theory is the independence of the RPMD rate coefficient on the choice of the dividing surface.

V. EXTENSION OF RING POLYMER RATE THEORY TO NONADIABATIC PROCESSES

The aim of the section is to develop an approximation to the full quantum nonadiabatic dynamics. For this purpose, we extend the RPMD method, which already takes into account tunneling and zero-point energy on a single adiabatic PES, to incorporate nonadiabatic transitions among different PESs via surface hopping. We outline below the mixed ring polymer surface-hopping (RPSH) dynamical scheme and, drawing upon analogies with the MQC rate theory of Sec. III C, we formulate an approximate approach for the computation of nonadiabatic quantum rates.

A. Ring polymer surface-hopping molecular dynamics

The extension of RPMD to nonadiabatic processes requires the formulation of the ring polymer electronic Hamiltonian. Based on physical arguments, we suggest two approximations. The centroid approximation (CA) results from the substitution of the classical nuclear position and momentum with the classical-like centroid position and momentum, respectively. In the bead approximation (BA), the ring polymer is interpreted as an effective "molecule" that moves on appropriately defined effective potential energy surfaces and each pair of surfaces is coupled by a single multidimensional nonadiabatic coupling vector. The approximations are discussed separately in Subsections V A 1 and V A 2.

1. Centroid approximation

As outlined in Sec. IV, in RPMD quantum particles are mapped on classical ring polymers whose centroids are equivalent to classical-like particles. We assume that the electronic Hamiltonian as well as the electronic states are parametrically dependent on time through the centroid trajectory, which takes the place of the classical nuclear trajectory. Then, the time-dependent Schrodinger equation that governs the evolution of the quantum subsystem becomes

$$i\dot{c}_i = V_i(\bar{R})c_i - i\sum_j \dot{\bar{R}} \cdot d_{ij}(\bar{R})c_j. \tag{35}$$

Equation (35) has the same form as Eq. (17) except that now it depends on the centroid position and velocity. As in the RPMD method, the centroid trajectory is obtained by the propagation of the whole ring polymer. The beads of the ring polymer move on a single adiabatic PES at all times except for instantaneous hops. During a hop all beads switch simultaneously to the new PES with their positions kept unchanged and the centroid momentum parallel to the nonadiabatic coupling vector is adjusted to conserve the total energy. If the required momentum reduction is greater than the component of the centroid momentum along the nonadiabatic coupling vector, the surface switch is rejected. The probability for a hop is determined by the FSSH criterion in Eq. (19) and the electronic amplitudes are calculated by the numerical integration of Eq. (35). This approximation closely resembles the FSSH method with the classical-like centroid trajectory of the ring polymer substituted for the classical nuclear motion.

2. Bead approximation

The centroid approximation to the electronic Hamiltonian in Eq. (35) is expected to be accurate at high temperatures when the beads of the ring polymer are mainly located close to the position of the centroid. With lowering the temperature the spatial extent of the ring polymer increases as the harmonic interactions between the beads are weakened. To capture the effect of delocalization, we suggest a modified form of the electronic Hamiltonian. We consider the ring polymer as an effective "molecule" with the beads being its constituent "atoms." Similar to atoms in a molecule, all beads reside on the same PES at all times; likewise, when an electronic transition occurs, the entire molecule (all of the beads) moves on the new PES. The adiabatic "molecular" PES is then the sum of two contributions: the average of the adiabatic PES over the bead positions and the harmonic interactions between adjacent beads. This definition corresponds to the effective potential of the ring polymer in the adiabatic RPMD Hamiltonian in Eq. (29). Further, the "molecular" nonadiabatic coupling vector is a single multidimensional vector for each pair of interacting electronic states with components represented by Eq. (18) for each "atom" or bead position, R_k. This "molecular" nonadiabatic coupling vector enters the electronic Hamiltonian through the dot product with the multidimensional vector of bead velocities. The resulting electronic Hamiltonian can be expressed as

$$i\dot{c}_i = \left[\frac{1}{N}\sum_{k=1}^{N} V_i(R_k)\right] c_i - i\sum_j \left[\frac{1}{N}\sum_{k=1}^{N} \dot{R}_k \cdot d_{ij}(R_k)\right] c_j. \tag{36}$$

The harmonic interactions have been omitted from the "molecular" PES in Eq. (36) as they do not contribute to the relative phase difference of the electronic amplitudes. Equation (36) represents the bead approximation to the electronic Hamiltonian. The details of the self-consistent propagation of ring polymers and electronic amplitudes remain essentially the same as in the CA approach with a single exception. To ensure proper energy conservation during a surface hop in the BA scheme, the adjustment of the bead momenta is made to the momentum component along the multidimensional "molecular" nonadiabatic coupling vector. Equation (36) readily reduces to the centroid form of Eq. (35) at high temperatures when the ring polymer shrinks substantially due to the increased stiffness of the harmonic interactions and the bead positions and velocities become approximately equal to the center-of-mass position and velocity.

B. Nonadiabatic ring polymer rate theory

With the aid of the RPSH method, the RPMD approximation to the flux-side correlation function in Eq. (30) can be generalized to nonadiabatic chemical processes. We pursue a similar extension as Eq. (10) to Eq. (20) in the classical case and obtain the following approximation to the quantum Kubo-transformed flux-side correlation function

$$C_{fs}^{\mathrm{RPSH}}(T,t) = \frac{1}{(2\pi)^N}\sum_i \iint d\mathbf{R}_0\, d\mathbf{P}_0 e^{-\beta_N H_N^{ad}(\mathbf{R}_0,\mathbf{P}_0;i)} \times \delta(\bar{R}_0 - R^{\neq})\frac{\bar{P}_0}{M} h(\bar{R}_t - R^{\neq}), \tag{37}$$

where

$$H_N^{ad}(\mathbf{P},\mathbf{R};i) = \sum_{k=1}^{N}\frac{P_k^2}{2M} + \sum_{k=1}^{N}\frac{M}{2}\omega_N^2(R_{k+1}-R_k)^2 + \sum_{k=1}^{N} V_i(R_k). \tag{38}$$

The rate coefficient can be calculated by

$$k_{\mathrm{RPSH}} = \frac{1}{Q_A^N}\lim_{t\to\infty} C_{fs}^{\mathrm{RPSH}}(T,t). \tag{39}$$

Here, Q_A^N is the quantum reactant partition function on multiple PESs in the asymptotic reactant region. As before, the dividing surface depends only on the centroid position of the ring polymer and is independent of the electronic state. The equilibrium distributions with centroids pinned at the dividing surface are sampled in the adiabatic approximation. The ring polymer positions and momenta, drawn in the canonical sampling, serve as initial conditions for the subsequent trajectory propagation at constant energy. As the dividing surface can often be located in the nonadiabatic coupling region, the trajectories are propagated by the RPSH method forward and backward in time. The rate coefficient is the average of the initial centroid momenta that belong to trajectories whose centroids start in the reactant region and end up in the product region after long enough time of propagation. Note that again the nonadiabatic effects are included in the dynamical evolution following the sampling of equilibrium initial conditions on each adiabatic PES. The particular computational algorithm is discussed in more detail in Sec. VI.

With obvious generalizations of Eqs. (24)–(26), the RPSH rate coefficient can be represented as

$$k_{\mathrm{RPSH}} = \frac{1}{Q_A^N}\sum_{in,fi}\sum_i C_{fs,i}^N(T,0_+)\kappa_{in,fi}^i, \tag{40}$$

where the state-selected transmission coefficients are now computed by the RPSH method. In Eq. (40) as before, the labels *in*, *i*, and *fi* designate the electronic state of the ring polymer trajectory in the reactant region, at the dividing surface and in the product region, respectively. The short-time limit of Eq. (40) is simply the centroid quantum transition state theory extended to multiple potential energy surfaces,

$$k_{\mathrm{MPES-QTST}} = \frac{1}{Q_A^N} \sum_i C_{fs,i}^N(T, 0_+) \tag{41}$$

with $C_{fs,i}^N(T, 0_+)$ the short-time limit of Eq. (30) on the *i*th PES.

Many of the attractive attributes of RPMD rate theory are preserved in the present extension to nonadiabatic processes. The rate expression gives the nonadiabatic classical rate theory of Sec. III C in the one-bead limit and the rate is approximately independent of the location of the dividing surface. Tunneling and zero-point energy are captured on each adiabatic PES and nonadiabatic quantum transitions are included by coherent, self-consistent propagation of ring polymers and electronic amplitudes. Note that, as with classical FSSH, decoherence of the electronic amplitudes can be included, if desired. We have not explored this here.

The RPSH method for the computation of approximate quantum nonadiabatic reaction rates is tested on a simple one-dimensional, two-state model and the results as well as a comparison with exact quantum-mechanical data are presented in Sec. VII.

VI. COMPUTATIONAL ALGORITHM

The computational algorithm for the calculation of the ring polymer nonadiabatic reaction rates consists of two stages that facilitate the numerical simulation of rare events.[49,50] In the first stage, the MPES-QTST rate coefficient of Eq. (41) is evaluated followed by the computation of the state-selected transmission coefficients, Eq. (40), in the second stage.

A. Transition state theory rate

The evaluation of MPES-QTST rates is performed efficiently by the definition of two dividing surfaces.[19,51] One of them, $\bar{R}_0^{\neq}$, is placed in the asymptotic reactant region, where in the considered model the MPES-QTST rate can be computed analytically. The position of the second dividing surface ($\bar{R}_1^{\neq}$) is adjusted variationally so as to minimize the thermal flux. The rates at the two surfaces are then connected by the reversible work needed to shift one of them into the other.

In particular, the MPES-QTST coefficient of Eq. (41) can be further rewritten as

$$k_{\mathrm{MPES-QTST}} = \sum_i k_i(\bar{R}_0^{\neq}) p_i(\bar{R}_1^{\neq}, \bar{R}_0^{\neq}), \tag{42}$$

where

$$k_i(\bar{R}_0^{\neq}) = \frac{1}{Q_A^N} C_{fs,i}^N(T, 0_+; \bar{R}_0^{\neq}) \tag{43}$$

and

$$p_i(\bar{R}_1^{\neq}, \bar{R}_0^{\neq}) = \frac{C_{fs,i}^N(T, 0_+; \bar{R}_1^{\neq})}{C_{fs,i}^N(T, 0_+; \bar{R}_0^{\neq})}. \tag{44}$$

Here, the dependence of the flux-side correlation function on the location of the dividing surface is denoted explicitly. For our one-dimensional model system the rate coefficient in Eq. (43) is simply

$$k_i(\bar{R}_0^{\neq}) = \frac{1}{2}(2\pi M\beta)^{-1/2}. \tag{45}$$

The factor of one-half accounts for the degeneracy of the electronic states in the asymptotic reactant region (see Sec. VII), while the second factor is the free particle thermal flux on a single PES. The thermal fluxes through the dividing surfaces in Eq. (44) can be easily singled out and the resulting expression can be put in a form facilitating numerical evaluation

$$p_i(\bar{R}_1^{\neq}, \bar{R}_0^{\neq}) = \frac{P_i(1)}{P_i(0)}, \tag{46}$$

where

$$P_i(\xi) = \frac{\iint d\mathbf{R}_0\, d\mathbf{P}_0 e^{-\beta_N H_N^{ad}(\mathbf{R}_0,\mathbf{P}_0;i)} \delta(\bar{R} - \bar{R}_\xi^{\neq})}{\iint d\mathbf{R}_0\, d\mathbf{P}_0 e^{-\beta_N H_N^{ad}(\mathbf{R}_0,\mathbf{P}_0;i)}} \tag{47}$$

is the thermal probability of the centroid position to take the value $\bar{R}_\xi^{\neq}$ with $0 \le \xi \le 1$. With the help of the centroid thermal distribution in Eq. (47), we can further define the potential of mean force

$$A_i(\xi) = -\frac{1}{\beta_N} \ln P_i(\xi). \tag{48}$$

Then, Eq. (46) takes the form

$$p_i(\bar{R}_1^{\neq}, \bar{R}_0^{\neq}) = e^{-\beta_N(A_i(1) - A_i(0))}. \tag{49}$$

The potential of mean force in Eq. (48) can be computed by a host of well-established techniques. In our implementation, we used the umbrella integration method of Kästner and Thiel.[52,53] The centroid reaction coordinate, $\bar{R}_\xi^{\neq}$, is divided in a number of windows in the range $0 \le \xi \le 1$ and a harmonic restraint potential is imposed in each window. The force constant of the umbrella potential is chosen large enough to keep the centroid in the vicinity of the center of the window but small enough to provide sufficient overlap between the separate window distributions. A separate ring polymer trajectory at constant temperature is evolved in each window and the potential of mean force is extracted from analysis of the first two moments of the centroid distributions in all windows. The ring polymer is propagated in the normal mode coordinates of the free ring polymer with masses of those modes shifted such that they evolve on a common time scale.[54] The temperature is maintained by a stochastic Langevin thermostat.[55] A separate calculation is performed for each potential energy surface and each temperature.

B. Transmission coefficients

The state-selected transmission coefficient of Eq. (40) is defined by

$$\kappa^i_{in,fi} = \frac{\lim_{t\to\infty} \iint d\mathbf{R}_0\, d\mathbf{P}_0 \delta_{in,n(-t)}\, e^{-\beta_N H_N^{ad}(\mathbf{R}_0,\mathbf{P}_0;i)} \delta(\bar{R}_0 - \bar{R}_1^{\neq}) \frac{\bar{P}_0}{M} h(\bar{R}_t - \bar{R}_1^{\neq})\, \delta_{fi,n(t)}}{C^N_{fs,i}(T, 0_+; \bar{R}_1^{\neq})} \tag{50}$$

with obvious generalization of Eq. (26). The evaluation of Eq. (50) proceeds in the following way. At time zero, the centroids of the ring polymers are pinned at the dividing surface located in the interaction region. The equilibrium distribution on the ith PES is sampled by running a constant-temperature ring polymer trajectory with the centroid motion eliminated. The constraint on the centroid is easiest to apply in the normal modes of the free ring polymer where the centroid coordinate decouples naturally from the higher normal coordinates. Besides the constraint on the centroid, the same constant-temperature integration algorithm can be used as in Sec. VI A. For each independent bead configuration, momenta are sampled at random from a Maxwell distribution. These positions and momenta then serve as initial conditions for the self-consistent propagation of the ring polymer and its electronic amplitudes first backward and then forward in time at constant energy. As the masses of the beads are now set at the physical mass of the quantum particle, the ring polymer motion is integrated by a multiple time step technique[56] alternating free ring polymer evolution and momenta updates with the forces arising from the PES. We use the 4th-order Runge-Kutta (RK4) method[57] to solve the time-dependent Schrodinger equation. The long-time step of the ring polymer integrator is used for the RK4 solver, and the matrix elements of the electronic Hamiltonian are updated with the aid of the positions and the velocities at the beginning, at the middle and at the end of the long-time step. Due to the large energy gap between the PESs in our model, the integration of the time-dependent Schrodinger equation is more stable in the interaction picture. For this purpose, we single out the adiabatic phases from the electronic amplitudes

$$\tilde{c}_i(t) = c_i(t) \exp\left\{ i \int_{-\infty}^{t} V_i(t')dt' \right\} \tag{51}$$

and Eq. (35) then becomes

$$\dot{\tilde{c}}_i = -\sum_j \dot{\bar{R}} \cdot d_{ij} \exp\left\{ -i \int_{-\infty}^{t} (V_j(t') - V_i(t'))dt' \right\} \tilde{c}_j. \tag{52}$$

Similar manipulations can be done with Eq. (36).

As the dividing surface is placed in the nonadiabatic coupling region, the correct initial conditions for the electronic amplitudes are not known. Moreover, the FSSH criterion depends on the history of the trajectory and consequently it cannot be applied if the electronic amplitudes are not properly initialized. Following Ref. 46, in the propagation backward in time we use a trial path-independent surface-hopping probability

$$f_{ij} = \frac{1}{2}\{1 - \exp(-|\dot{\bar{R}} \cdot d_{ij}|\Delta t)\} \tag{53}$$

and a similar expression for the BA approach. In addition, we evolve simultaneously two independent solutions of the Schrodinger equation. The latter are saved in the computer memory together with the matrix elements of the electronic Hamiltonian. In the asymptotic reactant region, the nonadiabatic coupling vanishes and the initial conditions for the electronic amplitudes are well defined

$$\begin{aligned} c_{in}(-\infty) &= 1, \\ c_{j\neq in}(-\infty) &= 0. \end{aligned} \tag{54}$$

The unique solution of the Schrodinger equation that satisfies Eq. (54) is then a linear combination of the time-evolved independent solutions obtained in the backward integration. The coefficients of the linear superposition are computed at the end of the backward propagation and the properly initialized electronic amplitudes are calculated at each time step along the backward trajectory. Now that the correct quantum amplitudes are known, the FSSH probability g_{ij} can be evaluated and the weight of the backward portion of the reactive trajectory can be adjusted according to the FSSH criterion of Eq. (19). Hence, we define a weighting factor W that is the product of the weighting contributions w_μ at each time step

$$W = \prod_\mu w_\mu. \tag{55}$$

If the ring polymer is in state i at the time step μ, then w_μ is defined as

$$w_\mu = g_{ij}/f_{ij},$$

if a switch to state j was attempted, and

$$w_\mu = \left(1 - \sum_{j\neq i} g_{ij}\right) \Big/ \left(1 - \sum_{j\neq i} f_{ij}\right), \tag{56}$$

if no switch was attempted. As a result of the backward integration, the initial conditions for the electronic amplitudes at the dividing surface can be found and the forward time evolution can proceed in the usual manner with the FSSH probability. The same computational algorithm is employed for both CA and BA approximations with the only difference being the momentum adjustment after a surface switch has occurred as discussed in Sec. V A.

Because we evolve trajectories backward and forward in time, the sampling over initial centroid momenta can be restricted to only positive outward momentum components from the dividing surface. This requires that the weight of each reactive trajectory is further reduced by the number of times it crosses the dividing surface in the forward direction. Finally, a large ensemble of RPSH trajectories is run and the state-selected transmission coefficient is the ratio of the averaged reactive flux to the total thermal flux at the dividing surface.

The calculation of the MQC rate coefficient follows similar procedure. In the classical case, the TST rate can be eval-

uated analytically as the model is one-dimensional, while the only difference in the computation of the transmission coefficients is that the ring polymer is substituted for a single bead ($N = 1$).

VII. COMPARISON WITH EXACT QUANTUM CALCULATIONS

In this section, we apply the method proposed above to a simple model problem that mimics a prototypical nonadiabatic bimolecular chemical reaction. The model is one-dimensional and has two electronic states. The mass of the quantum particle is chosen to be 2000 a.u., which is comparable to the mass of the hydrogen atom and leads us to expect significant quantum behavior. In the diabatic representation, the model potentials are

$$\begin{aligned} V_{11} &= A\exp(-BR^2), \\ V_{22} &= C\exp(2E(R-D)) + 2C\exp(E(R-D)), \\ V_{12} &= F\exp(-GR^2). \end{aligned} \tag{57}$$

The parameters for the present calculation are chosen to be $A = 5\times10^{-3}$, $B = 2$, $C = 1\times10^{-5}$, $D = -2.8$, $E = 1.2$, $F = 1\times10^{-3}$, and $G = 2$, all in atomic units. The potential energy curves in the adiabatic representation are obtained by diagonalization of the potential energy matrix and are shown in Fig. 1. In this figure, the nonadiabatic coupling strength computed by Eq. (18) is also shown. The ground state potential energy curve has a barrier, while the excited state curve is repulsive. The adiabatic states are degenerate along the asymptotic negative portion of the real axis. The model system resembles the PESs of the $F + H_2$ reaction,[58] but some of the physical parameters were intentionally changed to promote quantum behavior, thus making the model more challenging for the performance of the approximate quantum-mechanical approaches.

The dividing surface is placed at the top of the potential barrier on the ground potential energy curve. This is the best location that minimizes recrossing. The asymptotic reactant region extends to the left of the dividing surface. Similarly, the asymptotic product region is located to the right of the dividing surface along the positive real axis. The nonadiabatic coupling vanishes for large $|R|$. The total rate coefficients for this model reaction are calculated as described in the Appendix and the results are plotted in Fig. 2. Further, the total rate coefficients are decomposed into electronic state-selected contributions. The transmission rate coefficient represents the portion of the reactive flux that starts on the ground state PES in the asymptotic reactant region, whereas the transition rate coefficient is due to the contribution from the excited state population of the reactants.

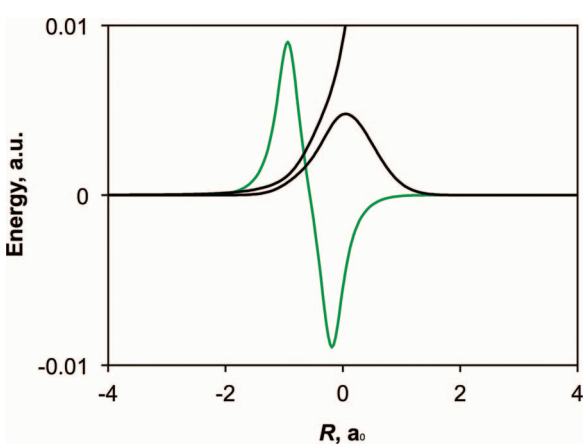

FIG. 1. Adiabatic potential energy curves (solid lines) and nonadiabatic coupling strength (green line, ×0.005) for the model system considered. All quantities are given in atomic units.

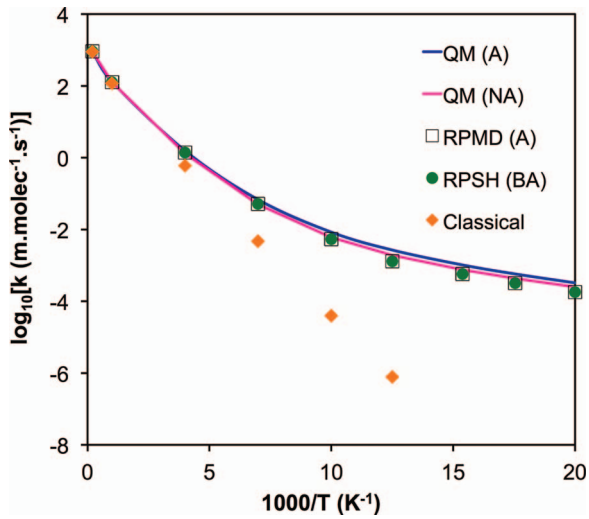

FIG. 2. Arrhenius plot of the numerically exact quantum adiabatic (blue line), quantum nonadiabatic (magenta line), ring polymer adiabatic (open squares), ring polymer surface-hopping in the BA (green circles), and classical surface-hopping (orange diamonds) total rate coefficients calculated in the temperature range between 50 K and 5000 K.

As shown in Fig. 2, the classical surface-hopping rate coefficients are orders of magnitude smaller than the quantum counterparts due to significant contribution of tunneling through the potential barrier on the ground PES. The ring polymer-based methods succeed in capturing the tunneling effect and provide quantitative agreement with the exact quantum estimates. On the other hand, the nonadiabatic coupling between the potential energy surfaces has a less pronounced influence and manifests itself in a small reduction of the total quantum rate compared to that evaluated in the adiabatic approximation. It can be shown[59] that the decrease in the total nonadiabatic rates relative to the adiabatic rates, especially in the low temperature regime, is due to the diagonal second derivative couplings.

It is shown in Figs. 3 and 4 that the RPSH technique apportions correctly the reactive flux between the reactant electronic states, unlike the adiabatic RPMD method, which assumes that the reaction takes place only on the ground PES. To quantify the latter observation, the relative difference of the ring polymer adiabatic and surface-hopping rate coefficients is defined as the difference of the two rates divided by the RPSH rate. As shown in Fig. 5, these relative differences reveal significant variation of the transmission rate coefficients with temperature, while the minus unity value of the transition contribution signifies the zero transition rate in the ring polymer adiabatic approach. Note also that a

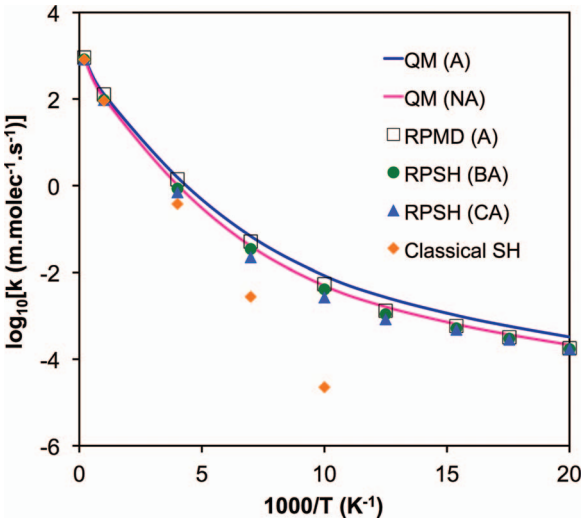

FIG. 3. Arrhenius plot of the numerically exact quantum adiabatic (blue line), quantum nonadiabatic (magenta line), ring polymer adiabatic (open squares), ring polymer surface-hopping, and classical surface-hopping (orange diamonds) transmission rate coefficients (initially on the ground state potential energy surface) in the same temperature range as in Fig. 2. The ring polymer surface-hopping rates are evaluated in the bead (green circles) and centroid (blue triangles) approximations.

mean-field-based method without an appropriate treatment of nuclear coherence or decoherence would fail to redistribute correctly the reactive flux between the electronic states. This is easy to justify: any admixture of the strongly repulsive (approaching positive infinity on the product side) excited PES in the averaged nuclear potential will necessarily lead to reflection with the result that the reactive flux vanishes in the asymptotic product region.

The evaluation of the electronic state-selected contributions to the total reaction rate provides a more stringent test

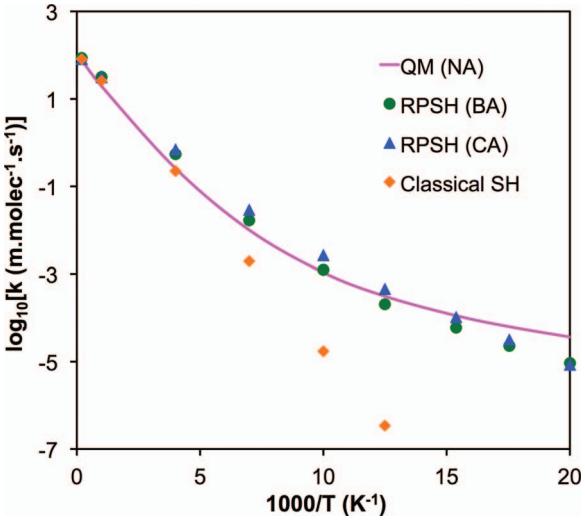

FIG. 4. Arrhenius plot of the numerically exact quantum nonadiabatic (magenta line), ring polymer surface-hopping, and classical surface-hopping (orange diamonds) transition rate coefficients (initially on the excited state potential energy surface) in the same temperature range as in Fig. 2. The ring polymer surface-hopping rates are evaluated in the bead (green circles) and centroid (blue triangles) approximations. Note that the transition rate coefficients vanish in the adiabatic approximation.

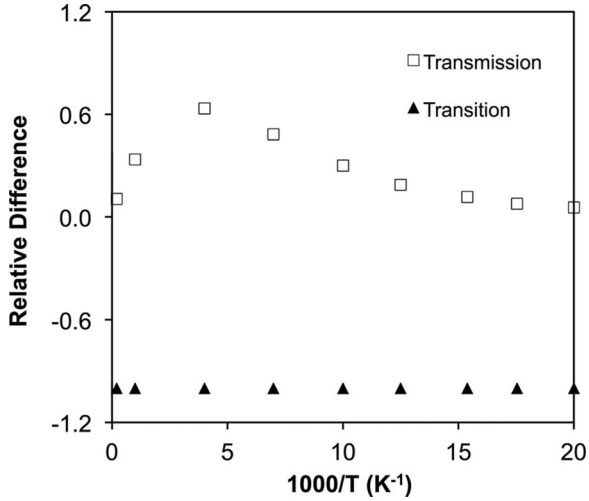

FIG. 5. Relative difference (defined in the text) of the ring polymer adiabatic and ring polymer surface-hopping (computed with the BA method) rate coefficients in transmission (open squares) and transition (black triangles) in the same temperature range as in Fig. 2. The value of minus unity of the transition contribution is due to the zero adiabatic RPMD transition rate coefficients.

of the performance of the present method. As expected, the adiabatic RPMD transmission rate underestimates the adiabatic quantum values throughout, while the behavior with respect to the nonadiabatic quantum rate is less uniform (Fig. 3). Both RPSH approximations succeed to take into account properly the nonadiabatic effects. The rates show slightly irregular behavior relative to the exact values being lower in the low temperature region and higher in the intermediate temperature regime. The latter can be attributed to the second derivative couplings, which decrease the total nonadiabatic rates as discussed before. Most importantly, the RPSH method allows the evaluation of the transition contribution to the total rate as shown in Fig. 4. In adiabatic RPMD, this contribution vanishes as the electronic states are uncoupled and there is no reactive flux on the excited state potential energy curve because it is repulsive. Overall, both the centroid and bead RPSH schemes provide good approximations to the quantum rates, with the latter performing somewhat better.

VIII. SUMMARY AND DISCUSSION

We have presented a method for the calculation of chemical rate constants in systems where both quantum statistical and nonadiabatic effects play a significant role. The method is practical and allows the simulation of chemical processes in systems with large numbers of atoms. Though considered in the light of reaction rates, the approach has potentially more general applicability to systems with quantum and near-classical degrees of freedom, where the motion of the latter needs to include quantum statistical effects. The method extends the RPMD approach to nonadiabatic processes and improves the classical dynamics of the nuclei in the FSSH scheme. This is achieved by a definition of a ring polymer electronic Hamiltonian, which determines the evolution of the electronic amplitudes through the time-dependent Schrodinger equation. Two approximations for the electronic Hamiltonian are proposed in which either the centroid

trajectory takes the place of the classical nuclear trajectory or the ring polymer is treated as an effective molecule. The surface-hopping scheme is an integral part of the approach. It assumes that all of the beads of a ring polymer reside on a single electronic state at all times except for instantaneous hops in which the beads switch states simultaneously. Both approximations to the ring polymer electronic Hamiltonian are tested against exact quantum mechanical calculations for a one-dimensional, two-state model system and the BA performs better than the CA especially in the low temperature regime. Since the computational expense of both approximations is comparable, we recommend the use of the BA for ring polymer surface-hopping simulations.

A major source of concern with the method is the particular surface-hopping scheme. As both RPMD and FSSH approaches are more or less *ad hoc* approximations in their realms, we have not provided a rigorous derivation of the proposed approach, but we have tried to justify it by drawing upon physical arguments. Yet, one can certainly envision other possibilities. For instance, a method could be developed in which the beads of the ring polymer hop independently among the PESs as opposed to hops of the whole ring polymer. But then, there would be a considerable chance that in regions where the nonadiabatic coupling vanishes, the beads evolve on different PESs – a picture that strongly resembles the Ehrenfest method[25,26] for mixed quantum-classical dynamics. The latter contradicts the principle underlying the surface-hopping approaches, i.e., classical particles always move on a single adiabatic PES. This is the guiding principle of our current method.

In the present extension of RPMD rate theory to nonadiabatic processes, we have approximated the exact quantum nonadiabatic equilibrium distribution by the adiabatic quantum Boltzmann distribution on multiple surfaces. In other words, we neglect ring polymer configurations with beads in a single necklace occupying different electronic states. The latter ring polymer configurations appear in the expression of the exact quantum nonadiabatic partition function and they contribute to the changes of the bead population density due to nonadiabatic effects at quantum thermal equilibrium.[60] As the RPSH approach is a mixed quantum-classical-like surface-hopping scheme, where the nuclear motion is described by the classical dynamics of ring polymers in an extended phase space, the RPSH method will share the equilibrium limit properties of the classical FSSH method, where nuclei simply follow classical trajectories. It has been shown[61] that the classical FSSH approach obeys detailed balance very closely, albeit not rigorously, and thus the FSSH dynamics recovers closely the equilibrium Boltzmann populations in the long-time limit. Therefore, it can be expected that the RPSH dynamics will very nearly satisfy detailed balance and hence it will approximately leave invariant the adiabatic quantum Boltzmann distribution on multiple surfaces. An essential requirement for a thermal correlation function to obey stationarity, i.e., independence on the initial time of observation, is the dynamics employed to leave invariant the underlying thermal distribution. Both classical FSSH and RPSH dynamics preserve approximately the thermal distributions in the respective classical and approximate quantum correlation functions, Eqs. (20) and (37), respectively, and hence the corresponding rate constant expressions are almost independent of the location of the dividing surface. These physical considerations need to be corroborated further by numerical evidence.

The RPSH dynamics of the present paper may be compared to the nonadiabatic path-integral transition state theory of Schwieters and Voth.[33,34] In the latter method, a single effective centroid potential energy surface is constructed by thermal averaging over both the nonzero normal modes of the ring polymer and all possible bead combinations among the electronic states. In contrast to our strategy, this scheme is essentially a mean-field approach in which the beads can redistribute separately among the electronic states thus creating a single electronic state-averaged potential energy surface. However, certain ideas might be borrowed from the latter approach for the refinement of our electronic Hamiltonian. For instance, one may envision that bead distributions among different electronic states can be included in an averaged effective way in the definition of the adiabatic potential energy surfaces and the multidimensional nonadiabatic coupling vector, resulting in higher order corrections to our electronic Hamiltonian. Nevertheless, this needs to be done in a consistent manner, which avoids double counting of nonadiabatic effects.

In the present formulation of the RPSH approach, the dephasing of the electronic amplitudes results from the divergence of the ring polymer trajectories and the averaging over the initial conditions. We have not attempted to include decoherence corrections in the surface-hopping algorithm. However, there are a number of prescriptions for decoherence corrections[62–65] to the traditional FSSH algorithm, which, if needed, can be applied to the present method as well.

In summary, based on the results of Sec. VII, the method developed here appears promising. It succeeds in capturing the tunneling and the correct distribution of reactants between the electronic states as seen from the quantitative agreement with the exact quantum results in Figs. 3 and 4. The influence of nonadiabaticity on the overall reaction rate constant for the model problem examined here is quite small, Fig. 2. However, the main success of the RPSH is shown in Fig. 4, the accurate branching ratio between the two electronic potential energy surfaces. Neither a mean-field-based method without nuclear coherence or decoherence incorporated nor adiabatic RPMD are able to provide even qualitatively correct reactive flux distribution between the electronic states. The method should be tested in more realistic multidimensional situations, where it can also be used for the estimation of kinetic isotope effects. In general, one-dimensional systems tend to exaggerate the deficiencies of approximate quantum-mechanical methods and we may expect that accuracy becomes higher in multidimensional systems. Nevertheless, the agreement with the accurate quantum-mechanical results is quite encouraging.

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APPENDIX: COMPUTATIONAL DETAILS

The quantum rate coefficients were obtained by numerical integration of the square of the constant energy scattering amplitudes. The latter were evaluated by exact propagation of wave packets with the split-operator Fourier transform method of Feit and Fleck[66] extended to multiple potential energy surfaces. A wave packet was initialized in the asymptotic reactant region and evolved forward in time until the scattering event was completed. An absorbing potential was placed in the asymptotic reactant region that eliminated the reflected portion of the scattered wave packet, while the transmitted component was projected out in the asymptotic product region. The scattering amplitudes were calculated by Fourier transform of the transmitted fraction of the wave packet and were properly normalized afterwards.

The rate coefficients were calculated for 9 different temperatures in the range between 50 K and 5000 K. The number of beads was 32 for $T \geq 143$ K and 128 for $T \leq 100$ K, respectively. The MPES-QTST rates were computed by umbrella integration with 100 windows placed in the range $-3a_0$ to $3a_0$ for the ground state PES and $-3a_0$ to $0.5a_0$ for the excited state PES, respectively. The force constant of the restraint harmonic potential was chosen to be 1.22 eV/a_0^2 after some experimentation and it is the same for all temperatures and all PESs. In each window, the ring polymer was equilibrated for 2.5×10^5 steps and data were sampled for the next 10^7 steps. The friction coefficient of the centroid mode was defined as twice the frequency of the umbrella potential. The time steps used for the simulations at different temperatures were: 1.2×10^{-2} fs for 50 K, 6×10^{-3} fs for 57 K and 65 K, 3.6×10^{-3} fs for 80 K and 100 K, 2.4×10^{-3} fs for 143 K and 250 K, 2.4×10^{-4} fs for 1000 K, and 6×10^{-5} fs for 5000 K.

The transmission coefficients were calculated with the rare event RPSH algorithm of Sec. VI. After initial equilibration of 10^6 steps, ring polymer configurations with centroids pinned at the dividing surface were sampled every 2.5×10^3 steps and momenta were drawn at random from Maxwell distribution. From each configuration and set of momenta a trajectory was run forward and backward in time. The total number of trajectories was 10^6 for each PES and temperature. The time steps were the same as those used in the canonical simulations except for the free ring polymer integration where they were reduced by 10 for $T \leq 250$ K, 50 for 1000 K, and 100 for 5000 K.

The classical transmission coefficients were evaluated by analogous procedure, where the number of beads in the ring polymer was set to one. For the calculation of the adiabatic RPMD transmission coefficients the ring polymer trajectories were propagated on the ground PES forward and backward in time. The parameters of the simulations were the same as those for the RPSH rate computation.

[1] M. Born and R. Oppenheimer, Ann. Phys. **389**, 457 (1927).
[2] E. Rabani, G. Krilov, and B. J. Berne, J. Chem. Phys. **112**, 2605 (2000).
[3] H. Wang, X. Sun, and W. H. Miller, J. Chem. Phys. **108**, 9726 (1998).
[4] I. R. Craig and D. E. Manolopoulos, J. Chem. Phys. **121**, 3368 (2004).
[5] I. R. Craig and D. E. Manolopoulos, J. Chem. Phys. **122**, 084106 (2005); **123**, 034102 (2005).
[6] T. F. Miller and D. E. Manolopoulos, J. Chem. Phys. **122**, 184503 (2005); **123**, 154504 (2005).
[7] J. Cao and G. A. Voth, J. Chem. Phys. **99**, 10070 (1993); **100**, 5093 (1994); **100**, 5106 (1994); **101**, 6157 (1994); **101**, 6168 (1994).
[8] S. Jang and G. A. Voth, J. Chem. Phys. **111**, 2357 (1999); **111**, 2371 (1999).
[9] E. Geva, Q. Shi, and G. A. Voth, J. Chem. Phys. **115**, 9209 (2001).
[10] W. H. Miller, Y. Zhao, M. Ceotto, and S. Yang, J. Chem. Phys. **119**, 1329 (2003).
[11] M. J. Gillan, Phys. Rev. Lett. **58**, 563 (1987).
[12] M. J. Gillan, J. Phys. C **20**, 3621 (1987).
[13] G. A. Voth, D. Chandler, and W. H. Miller, J. Chem. Phys. **91**, 7749 (1989).
[14] R. P. Feynman, A. R. Hibbs, and D. F. Styer, *Quantum Mechanics and Path Integrals: Emended Edition* (McGraw-Hill, New York, 2005).
[15] D. Chandler and P. G. Wolynes, J. Chem. Phys. **74**, 4078 (1981).
[16] R. Collepardo-Guevara, I. R. Craig, and D. E. Manolopoulos, J. Chem. Phys. **128**, 144502 (2008).
[17] A. R. Menzeleev, N. Ananth, and T. F. Miller, J. Chem. Phys. **135**, 074106 (2011).
[18] T. F. Miller III, J. Chem. Phys. **129**, 194502 (2008); A. R. Menzeleev and T. F. Miller III, *ibid.* **132**, 034106 (2010).
[19] R. Collepardo-Guevara, Y. V. Suleimanov, and D. E. Manolopoulos, J. Chem. Phys. **130**, 174713 (2009).
[20] Y. V. Suleimanov, R. Collepardo-Guevara, and D. E. Manolopoulos, J. Chem. Phys. **134**, 044131 (2011).
[21] J. C. Tully and R. K. Preston, J. Chem. Phys. **55**, 562 (1971).
[22] J. C. Tully, J. Chem. Phys. **93**, 1061 (1990).
[23] J. C. Tully, Int. J. Quantum Chem. **40**, 299 (1991).
[24] J. C. Tully, "Classical and quantum dynamics in condensed phase simulations," in *Proceedings of the International School of Physics LERICI, Villa Marigola, 1997*, edited by B. J. Berne, G. Ciccotti, and D. F. Coker (World Scientific, Singapore, 1998), p. 489.
[25] J. C. Tully, in *Modern Methods for Multidimensional Dynamics Computations in Chemistry*, edited by D. L. Thompson (World Scientific, Singapore, 1998), p. 34.
[26] J. C. Tully, Faraday Discuss. **110**, 407 (1998).
[27] X. Sun, H. Wang, and W. H. Miller, J. Chem. Phys. **109**, 7064 (1998).
[28] W. H. Miller, J. Phys. Chem. A **113**, 1405 (2009).
[29] M. Thoss and G. Stock, Phys. Rev. A **59**, 64 (1999).
[30] C. Zhu and H. Nakamura, J. Chem. Phys. **106**, 2599 (1997).
[31] H. Nakamura, J. Phys. Chem. A **110**, 10929 (2006).
[32] J.-L. Liao and G. A. Voth, J. Phys. Chem. B **106**, 8449 (2002).
[33] C. D. Schwieters and G. A. Voth, J. Chem. Phys. **108**, 1055 (1998).
[34] C. D. Schwieters and G. A. Voth, J. Chem. Phys. **111**, 2869 (1999).
[35] R. I. Cukier and D. G. Nocera, Ann. Rev. Phys. Chem. **49**, 337 (1998).
[36] S. Hammes-Schiffer and A. A. Stuchebrukhov, Chem. Rev. **110**, 6939 (2010).
[37] S. Hammes-Schiffer and J. C. Tully, J. Chem. Phys. **101**, 4657 (1994).
[38] J.-Y. Fang and S. Hammes-Schiffer, J. Chem. Phys. **106**, 8442 (1997); **107**, 8933 (1997).
[39] A. Nitzan, *Chemical Dynamics in Condensed Phases: Relaxation, Transfer, and Reactions in Condensed Molecular Systems* (Oxford University Press, New York, 2006).
[40] R. Kubo, J. Phys. Soc. Jpn. **12**, 570 (1957); R. Kubo, M. Yokota, and S. Nakajima, *ibid.* **12**, 1203 (1957).
[41] T. Yamamoto, J. Chem. Phys. **33**, 281 (1960).
[42] W. H. Miller, S. D. Schwartz, and J. W. Tromp, J. Chem. Phys. **79**, 4889 (1983).
[43] E. Wigner, Trans. Faraday Soc. **34**, 29 (1937); J. Chem. Phys. **5**, 720 (1937).
[44] P. Pechukas, Ann. Rev. Phys. Chem. **32**, 159 (1981).
[45] J. C. Keck, J. Chem. Phys. **32**, 1035 (1960).
[46] S. Hammes-Schiffer and J. C. Tully, J. Chem. Phys. **103**, 8528 (1995).
[47] J. O. Richardson and S. C. Althorpe, J. Chem. Phys. **131**, 214106 (2009).
[48] B. J. Braams and D. E. Manolopoulos, J. Chem. Phys. **125**, 124105 (2006).
[49] D. Frenkel and B. Smit, *Understanding Molecular Simulation* (Academic, San Diego, 2002), Chap. 16.
[50] D. Chandler, J. Chem. Phys. **68**, 2959 (1978).

[51]T. Yamamoto and W. H. Miller, J. Chem. Phys. **120**, 3086 (2004).
[52]J. Kästner and W. Thiel, J. Chem. Phys. **123**, 144104 (2005).
[53]J. Kästner and W. Thiel, J. Chem. Phys. **124**, 234106 (2006).
[54]M. E. Tuckerman, D. Marx, M. Klein, and M. Parrinello, J. Chem. Phys. **104**, 5579 (1996).
[55]M. Ceriotti, M. Parrinello, T. E. Markland, and D. E. Manolopoulos, J. Chem. Phys. **133**, 124104 (2010).
[56]M. Tuckerman, B. J. Berne, and G. J. Martyna, J. Chem. Phys. **97**, 1990 (1992).
[57]W. H. Press, S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery, *Numerical Recipes in Fortran 77: The Art of Scientific Computing* (Cambridge University Press, Cambridge, 1992).
[58]J. C. Tully, J. Chem. Phys. **60**, 3042 (1974).
[59]Unpublished results.
[60]J. R. Schmidt and J. C. Tully, J. Chem. Phys. **127**, 094103 (2007).
[61]P. V. Parandekar and J. C. Tully, J. Chem. Phys. **122**, 094102 (2005); J. R. Schmidt, P. V. Parandekar, and J. C. Tully, *ibid.* **129**, 044104 (2008).
[62]J. E. Subotnik and N. Shenvi, J. Chem. Phys. **134**, 024105 (2011).
[63]C. Zhu, S. Nangia, A. W. Jasper, and D. G. Truhlar, J. Chem. Phys. **121**, 7658 (2004).
[64]S. C. Cheng, C. Zhu, K. K. Liang, S. H. Lin, and D. G. Truhlar, J. Chem. Phys. **129**, 024112 (2008).
[65]G. Granucci, M. Persico, and A. Zoccante, J. Chem. Phys. **133**, 134111 (2010).
[66]M. D. Feit, J. A. Fleck, Jr., and A. Steiger, J. Comput. Phys. **47**, 412 (1982).